Chapter 4: Electrostatics

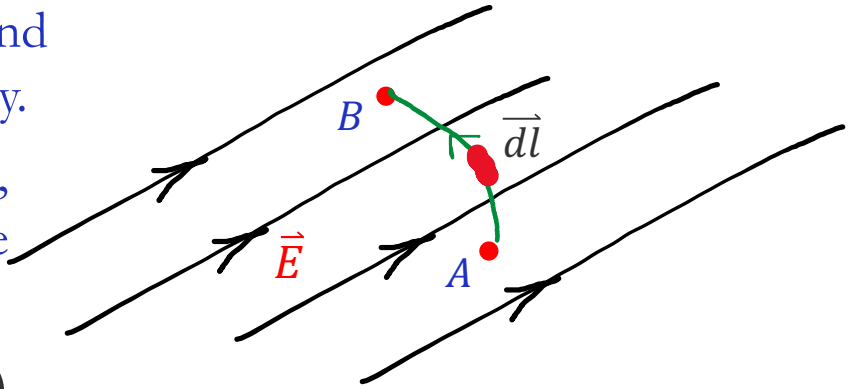
The following subjects will be discussed

- Definition of the Electric Potential.
- Potential-Field Relations.
- Potential Calculations for Charge Distributions.
- Energy Density in Electrostatic Fields.

The Electric Potential

The electric field can also be related and evaluated from the concept of potential energy.

Starting with the electrostatic force relation, we can find the work done on a charge moving within some electric field distribution.



$$\vec{F} = q\vec{E} \quad \rightarrow \quad dW = -\vec{F}\cdot\vec{dl} = -q\left(\vec{E}\cdot\vec{dl}\right)$$

The total work done by the field (energy gained or lost), going from point A to point B, is

$$W = -q\int_A^B \vec{E}\cdot\vec{dl}$$

The electric potential between two points (A and B) is defined as the amount of work per unit charge needed to bring a charge (q) from point A to point B within a field $\vec{E}$.

$$V_{AB} = W/q = -\int_A^B \vec{E}\cdot\vec{dl}$$

Because the electric force is conservative,

$$V_{AB} = -V_{BA} \rightarrow V_{AB} + V_{BA} = 0 \quad \rightarrow \quad \oint \vec{E}\cdot\vec{dl} = 0 \quad \rightarrow \quad \vec{\nabla}\times\vec{E} = 0 \quad \rightarrow \quad \vec{E} = -\vec{\nabla}V$$

Stokes Theorem

The rate of change in potential speaks about the properties of the field.

Example

A proton is accelerated by the field in the set-up given below from rest.

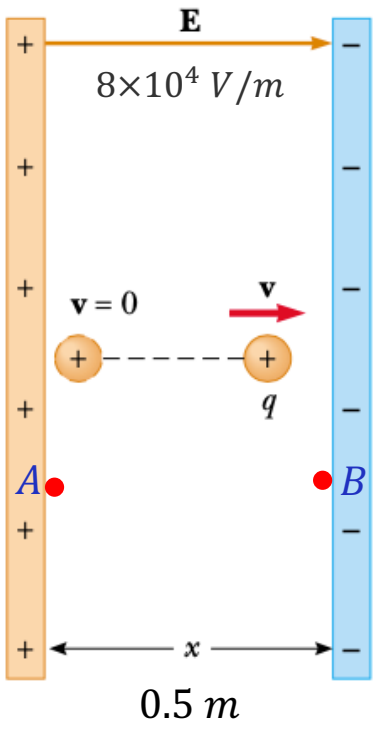
1) Find the potential difference $V_B - V_A$ and $V_A - V_B$.

$$V_B - V_A = -Ed = -8\times10^4\,(0.5) = -4\times10^4 \text{ (V)}$$

(constant field)

(A is at a higher potential than B)

$$V_A - V_B = -(V_B - V_A) = 4\times10^4 \text{ (V)}$$



2) As the proton travels from A to B, what is the loss/gain in the system energy?

$$Loss = \Delta U = (V_B - V_A)\,q = -4\times10^4\times1.6\times10^{-19} = -6.4\times10^{-15} \text{ (J)}$$

$$Loss = \frac{-6.4\times10^{-15}}{1.6\times10^{-19}} = -4\times10^4 \text{ (eV)}$$

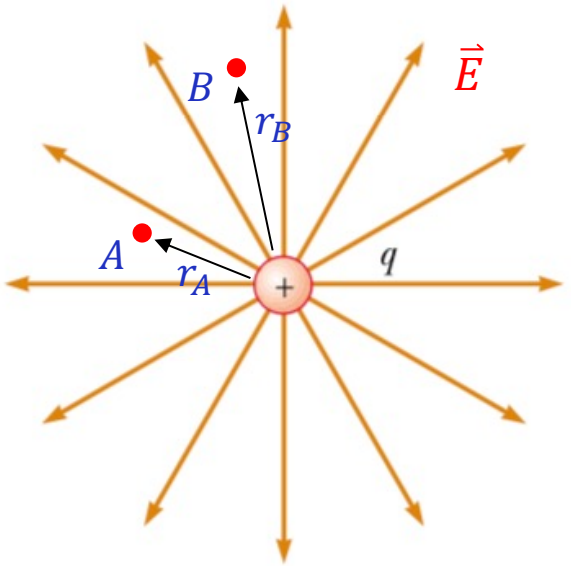
(This energy loss is gained by the proton as KE)

Potential Calculations: A Point Charge

The field-potential relation can be used to calculate the potential difference between any two points. Starting with the field equation, the potential difference can be found using

$$V_B - V_A = -\int_A^B \vec{E} \cdot \overrightarrow{dl}$$

Let's apply it for a point charge case. The electric field due to a point charge is given by $\vec{E} = \frac{1}{4\pi\varepsilon_o}\frac{q}{r^2}\,\vec{a}_r$



$$V_B - V_A = -\frac{1}{4\pi\varepsilon_o}\int_A^B \frac{q}{r^2}\,\vec{a}_r \cdot dr\,\vec{a}_r = -\frac{q}{4\pi\varepsilon_o}\int_{r_A}^{r_B}\frac{1}{r^2}\,dr$$

$$V_B - V_A = -\frac{q}{4\pi\varepsilon_o}\left.\frac{1}{r}\right|_{r_A}^{r_B} = \frac{q}{4\pi\varepsilon_o}\left[\frac{1}{r_B} - \frac{1}{r_A}\right]$$

$$V_B - V_A = \frac{q}{4\pi\varepsilon_o}\left[\frac{1}{r_B} - \frac{1}{r_A}\right]$$ Potential difference in the space of a point charge

$$V_p = \frac{q}{4\pi\varepsilon_o}\left[\frac{1}{r_p}\right]$$ Potential at a point

Remarks

1) Potential behaves as $1/r$.
2) If we take the reference as $V(\infty) = 0$, then we can define the potential at a point independently.

Potential Calculations: Many Point Charges

For many point charges, use superposition to find the potential at a point in the space of the charges.

$$V_p = \frac{1}{4\pi\varepsilon_o}\sum \frac{q_i}{r_i}$$

Example

Consider two charges, as follows.

q_1=2 μC at (0, 0)

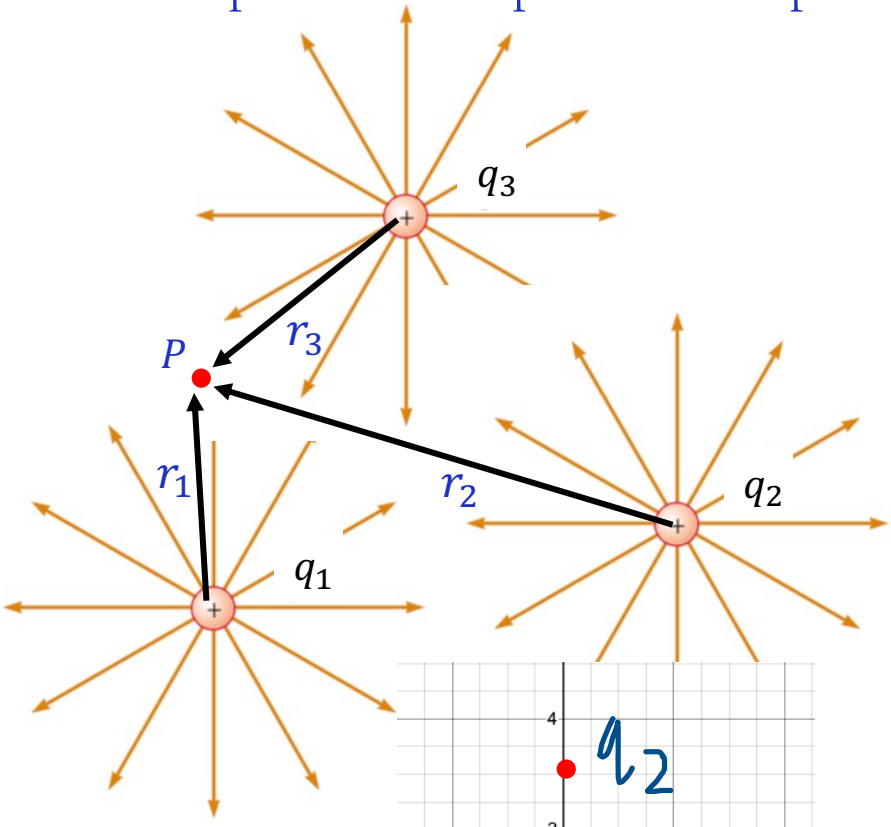
q_2=-6 μC at (0, 3 m)

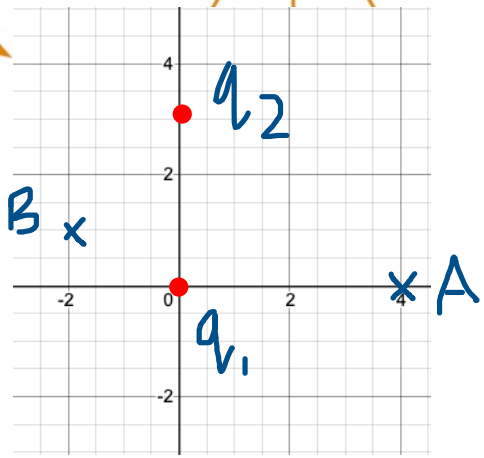
1) Find the potential at A(4 m, 0).

Take V(∞)=0.

$$V_A = \frac{1}{4\pi\varepsilon_0}\left[\frac{2\mu}{4} + \frac{-6\mu}{5}\right] = -6.29 \text{ kV}$$

2) Find the potential difference between A and B(-2 m, 1 m), V_{AB}.





Potential Calculations: Charge Distributions

In the case of continuous charge distributions, defined by a charge density (ρ_L, ρ_s or ρ_v), the electric potential is calculated using an integral equation.

Starting with the point charge equation

$$V = \frac{q}{4\pi\varepsilon\, r} \quad \rightarrow \quad dV = \frac{dq}{4\pi\varepsilon\, r}$$

$$V = \frac{1}{4\pi\varepsilon}\int \frac{dq}{r} = \frac{1}{4\pi\varepsilon}\int \frac{\rho_L}{r}\, dl$$ Line charge

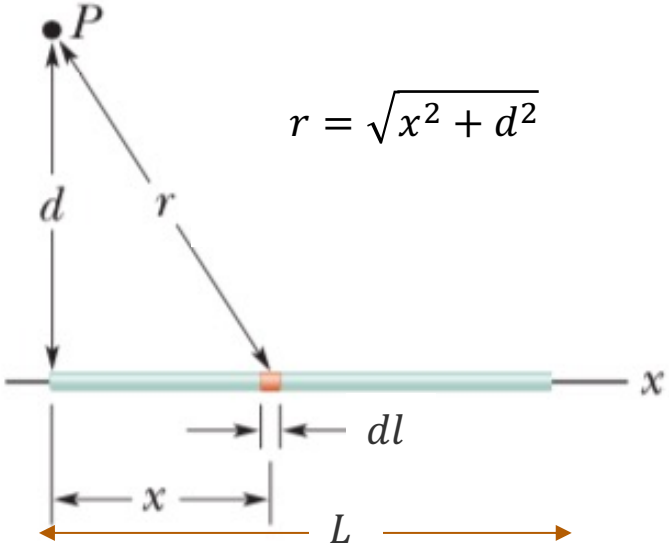
$$V = \frac{1}{4\pi\varepsilon}\int \frac{dq}{r} = \frac{1}{4\pi\varepsilon}\int \frac{\rho_s}{r}\, dS$$ Surface charge

$$V = \frac{1}{4\pi\varepsilon}\int \frac{dq}{r} = \frac{1}{4\pi\varepsilon}\int \frac{\rho_v}{r}\, dv$$ Volume charge

Potential Calculations: Charge Distributions
Line

Find the electric potential at any point due to a thin line charge (continuous distribution). The line has a uniform charge density ρ_L (C/m).

The point P is directly above the left end of the line. The final result will only be good for this case. But it also applies for the right end of the line.



For a very small length of the line dl, we can write

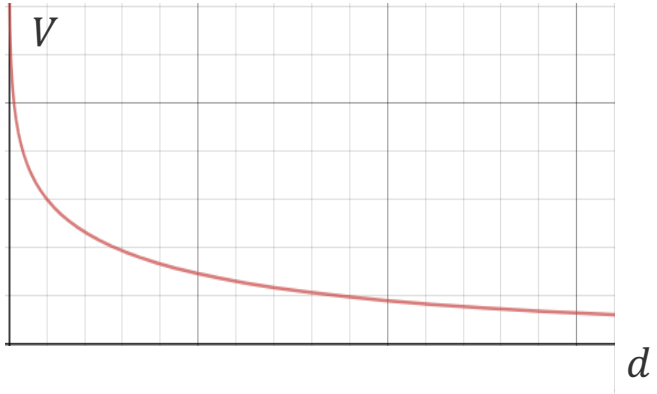
$$dV_P = \frac{dQ}{4\pi\varepsilon_0 r} \qquad dQ = \rho_L dl = \rho_L dx$$

Notice that we will integrate over the whole length of the line. So, the differential in the integration will be dx. Let's write all other variables in terms of x.

$$V_P = \frac{\rho_L}{4\pi\varepsilon_0}\int_0^L \frac{1}{\sqrt{x^2+d^2}}\,dx$$

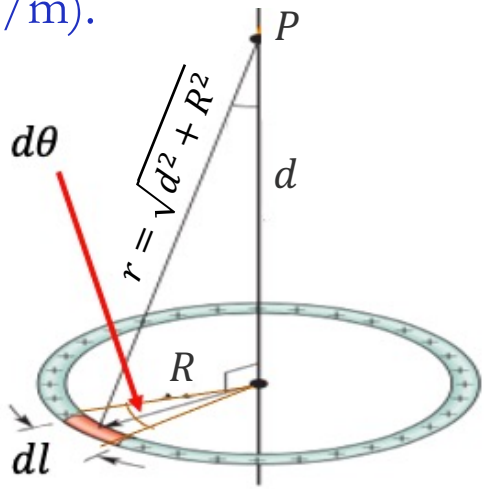
Use trigonometric substitution (put $x = d\,tan\theta$)

$$V_P = \frac{\rho_L}{4\pi\varepsilon_0}\ln\left\{\frac{L+\sqrt{L^2+d^2}}{d}\right\}$$



Potential Calculations: Charge Distributions Line

Find the electric potential at any point due to a thin ring charge (continuous distribution). The ring has a uniform charge density ρ_L (C/m).



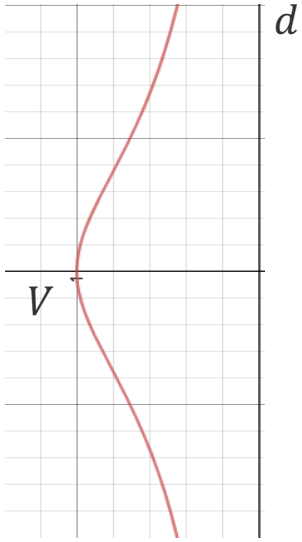
The point P is along the axis of the ring. The final result will only be good for this case.

For a very small length of the ring dl, we can write

$$dV_P = \frac{dQ}{4\pi\varepsilon_0 r} \qquad dQ = \rho_L dl = \rho_L R\, d\theta$$

Notice that we will integrate over the whole length of the ring. So, the differential in the integration will be $d\theta$. Let's write all other variables in terms of θ.

$$V_P = \frac{\rho_L R}{4\pi\varepsilon_0\sqrt{d^2+R^2}} \int_0^{2\pi} d\theta \quad \Rightarrow \quad V_P = \frac{\rho_L R}{2\varepsilon_0\sqrt{d^2+R^2}}$$



The potential is NOT zero at the center of the ring, (in fact, it's maximum).
Why?

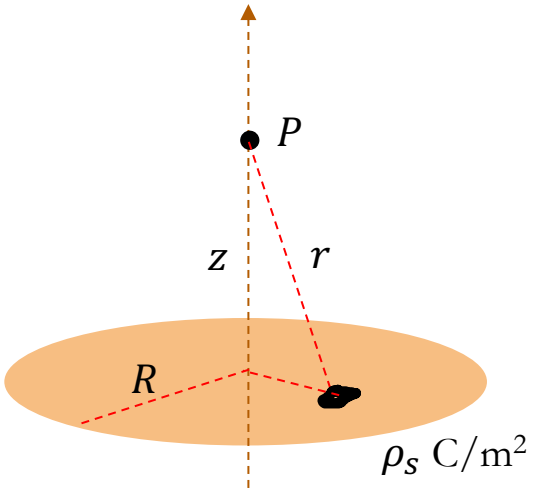
Potential Calculations: Charge Distributions Disk

Find the electric potential at any point due to a thin disk charge (continuous distribution). The disk has a uniform charge density ρ_s (C/m^2), radius R.

The point P is along the axis of the disk. The final result will only be good for this case.

$$V = \frac{1}{4\pi\varepsilon}\int \frac{\rho_s}{r}\, dS \qquad dS = \rho\, d\rho\, d\varphi \qquad r = \sqrt{\rho^2 + z}$$

$$V = \frac{\rho_s}{4\pi\varepsilon}\int_0^{2\pi}\int_0^{R} \frac{\rho}{\sqrt{\rho^2 + z^2}}\, d\rho\, d\varphi = \frac{\rho_s}{2\varepsilon}\int_0^{R} \frac{\rho}{\sqrt{\rho^2 + z^2}}\, d\rho$$

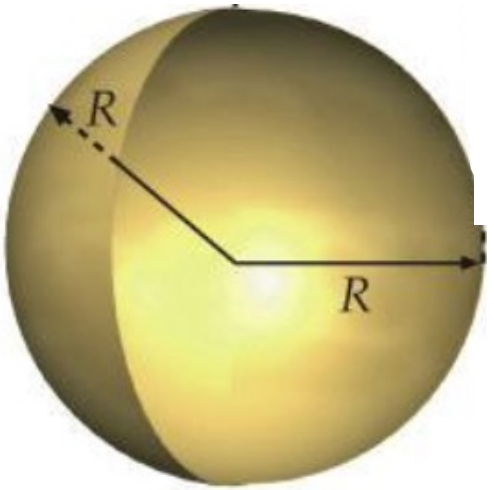


Use substitution
($u = z^2 + \rho^2$)

$$V_P = \frac{\rho_s}{2\varepsilon}\left(\sqrt{z^2 + R^2} - z\right)$$

Potential Calculations: Charge Distributions Sphere

Find the electric potential at any point due to a charged solid non-conducting sphere. The sphere has a uniform charge density $\rho_v (\mathrm{C/m^3})$, radius R.



Find the potential at any point inside or outside the sphere.

We can use the same procedure as before by taking a very small volume dv and finding the charge dQ.

An alternative method is to find the potential from the electric field using the following equation.

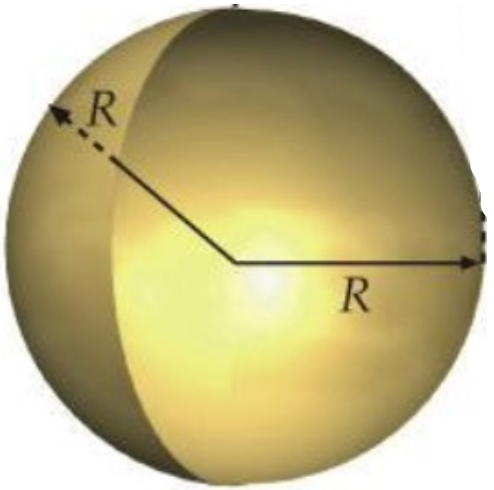
$$V_P = -\int \vec{E} \cdot \overrightarrow{dl}$$

Notice that this is an indefinite integral. So, a constant of integration will appear in the final answer.

Do we know the electric field everywhere for this sphere case? Yes. We have done this problem as an application of Gauss' Law (Check your notes).

Potential Calculations: Charge Distributions Sphere

Find the electric potential at any point due to a charged solid non-conducting sphere. The sphere has a uniform charge density ρ_v(C/m^3), radius R.



1) $r \leq R$ (points inside sphere)

$$\vec{E} = \frac{\rho_v\, r}{3\varepsilon_0}\,\vec{a}_r \quad \Rightarrow \quad V = -\int \vec{E}\cdot\vec{dl} = -\frac{\rho_v}{3\varepsilon_0}\int r\, dr$$

$$V = -\frac{\rho\, r^2}{6\varepsilon_0} + C_1$$

2) $r \geq R$ (points outside sphere)

$$\vec{E} = \frac{\rho_v R^3}{3\varepsilon_0 r^2}\,\vec{a}_r \quad \Rightarrow \quad V = -\int \vec{E}\cdot\vec{dl} = -\frac{\rho_v\, R^3}{3\varepsilon_0}\int \frac{1}{r^2}\, dr$$

$$V = \frac{\rho_v\, R^3}{3\varepsilon_0}\,\frac{1}{r} + C_2$$

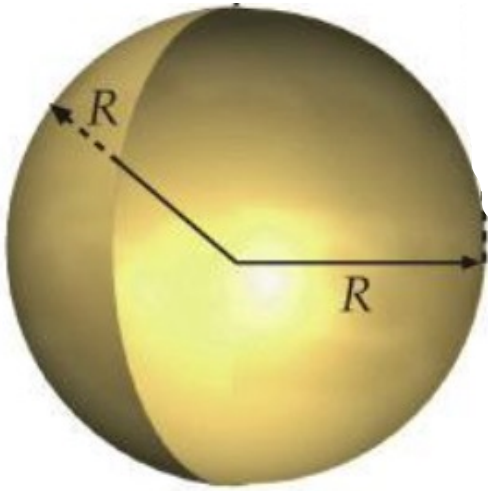
To evaluate the constants of integration, we use boundary conditions; two constants means we need two boundary conditions.

1) V(∞)=0.

2) At $r = R$, the two solutions are equal.

Potential Calculations: Charge Distributions Sphere

Find the electric potential at any point due to a charged solid non-conducting sphere. The sphere has a uniform charge density ρ_v (C/m^3), radius R.



Applying $V(\infty) = 0$ condition means a point outside sphere

$$V(\infty) = \frac{\rho_v R^3}{3\varepsilon_0}\frac{1}{\infty} + C_2 = 0 \qquad C_2 = 0$$

Applying $V(R)$ condition

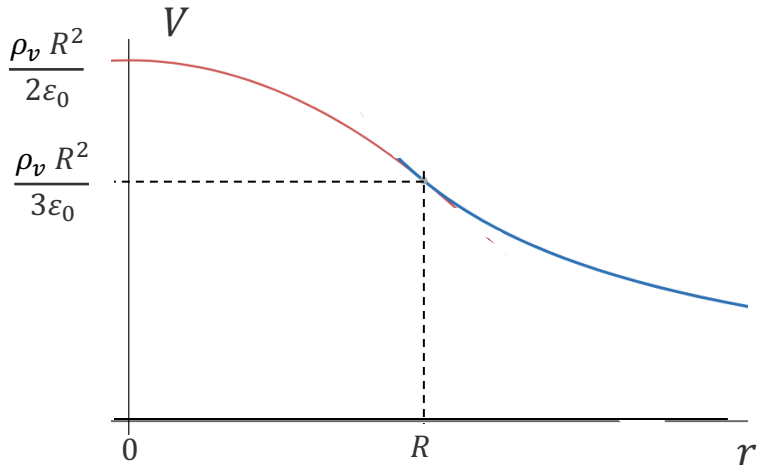
$$\frac{\rho_v R^3}{3\varepsilon_0}\frac{1}{R} = -\frac{\rho_v R^2}{6\varepsilon_0} + C_1 \qquad C_1 = \frac{\rho_v R^2}{2\varepsilon_0}$$

1) $r \le R$ (points inside sphere)

$$V = \frac{\rho_v}{6\,\varepsilon_0}(3R^2 - r^2)$$

2) $r \ge R$ (points outside sphere)

$$V = \frac{\rho_v R^3}{3\varepsilon_0}\frac{1}{r}$$

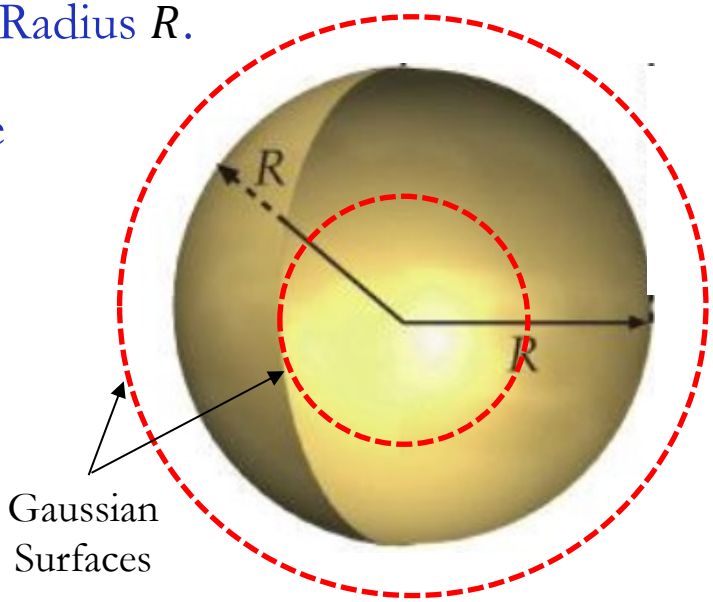


Potential Calculations: Charge Distributions Shell

Find the electric potential at any point due to a conducting shell. The shell has a total charge of Q distributed uniformly over the surface. Radius R.

Find the potential at any point inside or outside the shell.

Here we can use the same approach, because we can easily find the electric field using Gauss' Law.



$$Q_{enclosed} = \varepsilon_0 \int \vec{E} \cdot \overrightarrow{dA}$$

For the left-hand side

We have two regions:

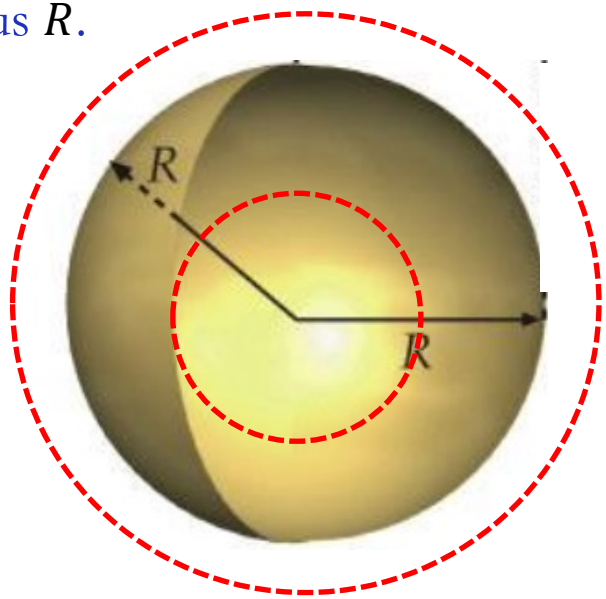
1) $r \leq R$: The total $Q_{enclosed}$ is zero.
2) $r \geq R$: The total $Q_{enclosed}$ is Q.

For the right-hand side

In both regions, E is constant over any spherical Gaussian surface so it goes out of the integral. The integral gives the surface area of the Gaussian surface $4\pi r^2$.

Potential Calculations: Charge Distributions Shell

Find the electric potential at any point due to a conducting shell. The shell has a total charge of Q distributed uniformly over the surface. Radius R.



1) $r \leq R$ (points inside sphere)

$$Q_{enclosed} = 0 \Rightarrow \vec{E} = 0 \Rightarrow V = -\int \vec{E} \cdot \overrightarrow{dl} = C_1$$

2) $r \geq R$ (points outside sphere)

$$Q_{enclosed} = Q \Rightarrow \vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\,\vec{a}_r$$

$$V = -\int \vec{E} \cdot \overrightarrow{dl} = -\frac{Q}{4\pi\varepsilon_0}\int \frac{1}{r^2}\,dr \Rightarrow V = \frac{Q}{4\pi\varepsilon_0}\frac{1}{r} + C_2$$

To evaluate the constants of integration, we use boundary conditions; two constants means we need two boundary conditions.

1) V(∞)=0.

2) At $r = R$, the two solutions are equal.

Potential Calculations: Charge Distributions Shell

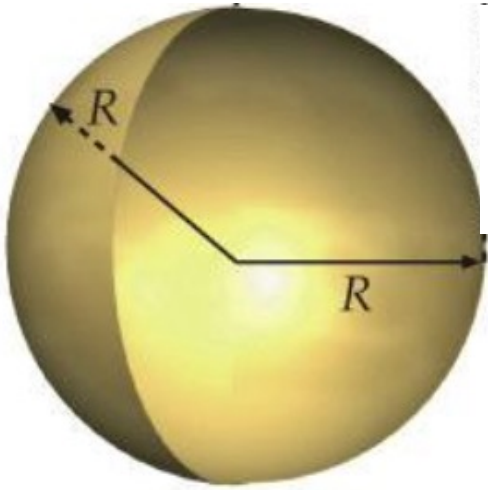
Find the electric potential at any point due to a conducting shell. The shell has a total charge of Q distributed uniformly over the surface. Radius R.

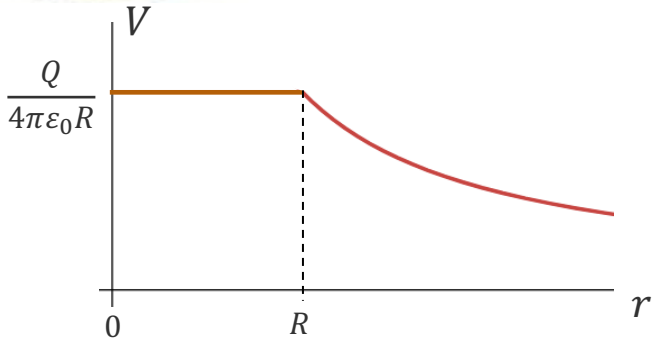
Applying $V(\infty) = 0$ condition means a point outside sphere

$$V(\infty) = \frac{Q}{4\pi\varepsilon_0}\frac{1}{\infty} + C_2 = 0 \qquad C_2 = 0$$

Applying $V(R)$ condition

$$\frac{Q}{4\pi\varepsilon_0}\frac{1}{R} = C_1 \qquad C_1 = \frac{Q}{4\pi\varepsilon_0 R}$$





1) $r \leq R$ (points inside sphere)

$$V = \frac{Q}{4\pi\varepsilon_0}\frac{1}{R}$$

2) $r \geq R$ (points outside sphere)

$$V = \frac{Q}{4\pi\varepsilon_0}\frac{1}{r}$$

Notice that because the electric field is zero inside the sphere, the potential is constant. All points inside the sphere, including the surface, are at the same potential.

Potential Calculations: Summary

$$dV = \frac{dq}{4\pi\varepsilon_0 r}$$

Approach (1)

Start from basics. Find a differential potential. Set-up the integral. Integrate to find potential expression.

$$V = -\int \vec{E} \cdot \vec{dl}$$

Approach (2)

Start from the electric field expressions (if known), or find electric field if easy. Use the potential-field relation to find the potential expression.

Examples

(1) The potential function in space is given by the following expression (which is the potential along the axis of the disk we studied before, slide 11)

$$V(z) = \frac{\rho_s}{2\varepsilon_o}(\sqrt{z^2 + R^2} - z)$$

Find the corresponding electric field.

We notice that the potential is only a function of z. So,

$$\vec{E}(z) = -\nabla V = -\frac{dV}{dz}\vec{a}_z$$

$$\vec{E}(z) = \frac{\rho_s}{2\varepsilon_o}\left[1 - \frac{z}{\sqrt{z^2 + R^2}}\right]\vec{a}_z$$

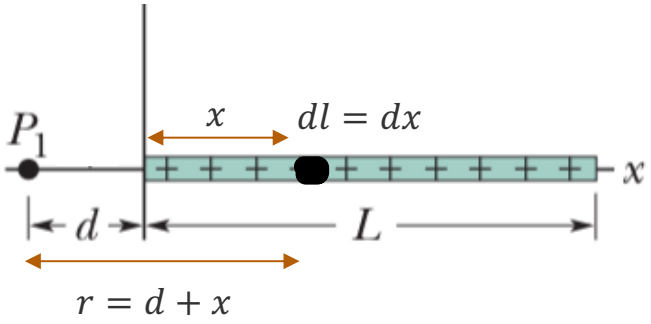
It is the same result that we obtained when we studied the electric field subject.

Examples

(2) A very thin finite line with uniform charge density ρ_L (C/m) and length L, placed along the x-axis, as in the figure.

Find an expression for the electric potential at any point along the negative x-axis.

$$dV_P = \frac{dq}{4\pi\varepsilon_0 r} \qquad dq = \rho_L\, dl = \rho_L\, dx$$



$$V_P = \frac{\rho_L}{4\pi\varepsilon_0}\int_0^L \frac{1}{d+x}\, dx$$

$$V_P = \frac{\rho_L}{4\pi\varepsilon_0}\ln\left\{\frac{d+L}{d}\right\}$$

Examples

(3) Consider a very thin infinite line with uniform charge density $\rho_L = 1$ (nC/m) is placed along the y-axis, as in the figure.

Find the potential difference V_{AB}, where $A(-3, 2, 4)$ and $B(6, 1, 0)$.

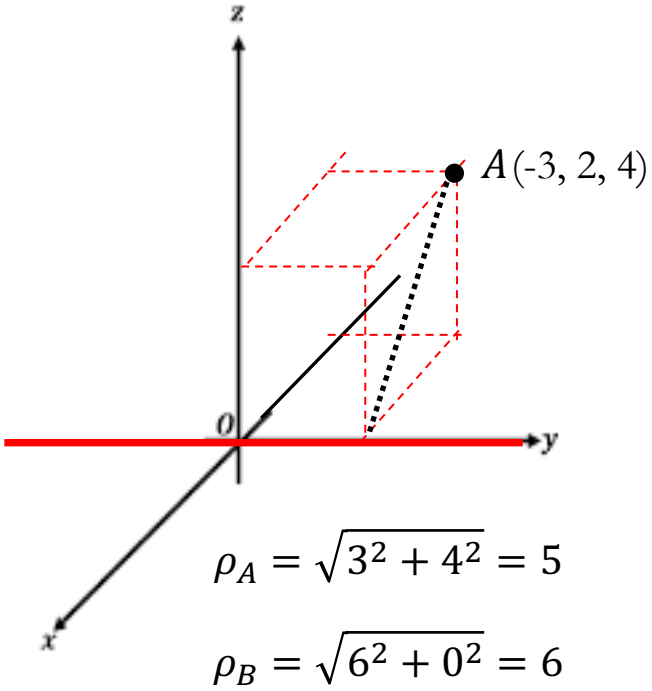
Starting with $\vec{E}_p = \frac{\rho_L}{2\pi\varepsilon\, s}\vec{a}_n$

$$V_{AB} = \int_A^B \vec{E}\cdot\vec{dl} = -\frac{\rho_L}{2\pi\varepsilon}\int_A^B \frac{1}{\rho}\, d\rho = -\frac{\rho_L}{2\pi\varepsilon}\ln\rho\Big|_{\rho_A}^{\rho_B}$$

$$V_{AB} = \frac{\rho_L}{2\pi\varepsilon}\ln\left(\frac{5}{6}\right) = -3.3\ V$$

Given $V_A = 5\ V$, then

$$V_{AB} = V_B - V_A = -3.3\ V \quad \rightarrow \quad V_B = 1.7\ V$$



$$\rho_A = \sqrt{3^2 + 4^2} = 5$$

$$\rho_B = \sqrt{6^2 + 0^2} = 6$$

Stored Electric Potential Energy

Consider a system consisting of three particle charges. We would like to find the stored electric potential energy in this system.

Take $W_{initial} = 0$

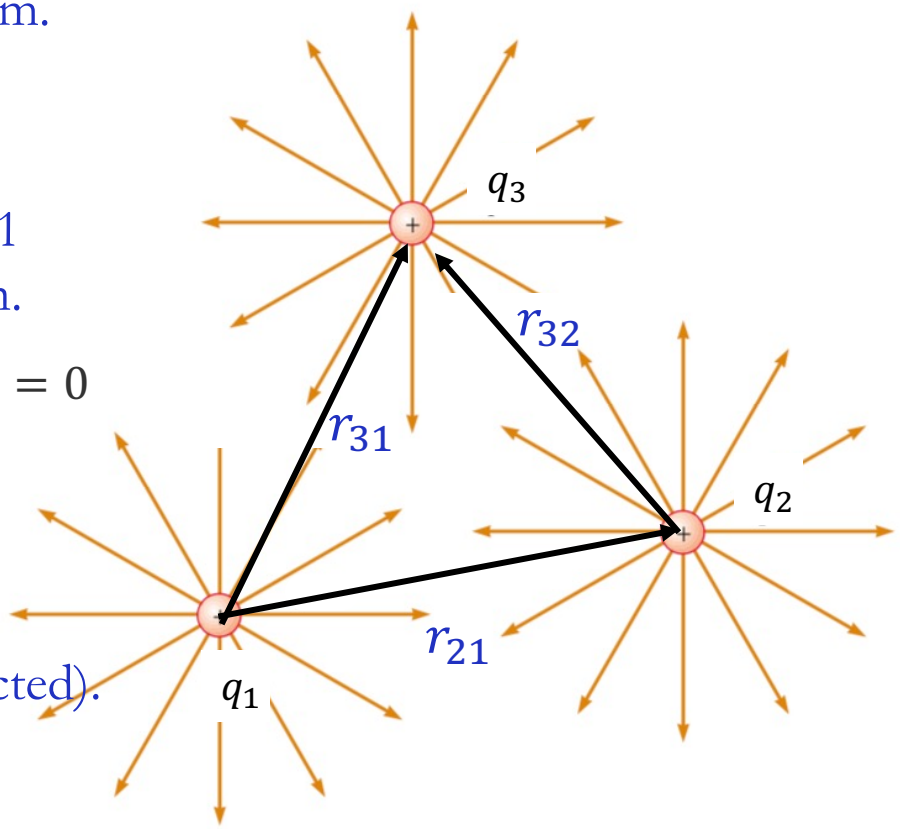
Assume that the system is assembled by bringing one charge at a time starting with 1 then 2 then 3, each one to its given location.

Bring q_1, no change in potential energy. $W_1 = 0$

Bring q_2, potential energy is added (subtracted). $W_2 = q_2 V_1 = q_2 \frac{q_1}{4\pi\varepsilon_o} \frac{1}{r_{21}}$

Bring q_3, potential energy is added (subtracted).

$$W_3 = q_3 V_1 + q_3 V_2 = q_3 \frac{q_1}{4\pi\varepsilon_o} \frac{1}{r_{31}} + q_3 \frac{q_2}{4\pi\varepsilon_o} \frac{1}{r_{32}}$$



Stored electric potential energy

$$W_{tot} = W_1 + W_2 + W_3 = \frac{q_1 q_2}{4\pi\varepsilon_o r_{21}} + \frac{q_1 q_3}{4\pi\varepsilon_o r_{31}} + \frac{q_2 q_3}{4\pi\varepsilon_o r_{32}}$$

$$W_E = \frac{1}{2}\sum q_i V_i$$

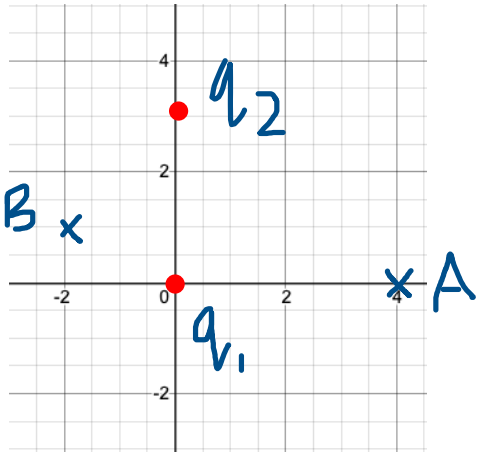
Stored Electric Potential Energy

Example

Consider two charges, as follows.

q_1=2 μC at (0, 0)

q_2=-6 μC at (0, 3 m)



3) What is the change in system potential energy if a charge q_3=3 μC is brought to point A?

$$\Delta U = q_3 V_A = (3\ \mu)(-6.29\ k) = -18.9\ mJ$$

The system lost potential energy because the field is "helping" the third charge.

4) If all three charges were brought (in order) from infinity to their respective locations, what is the change in system electric potential energy?

$$\Delta U = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1 q_2}{3} + \frac{q_1 q_3}{4} + \frac{q_2 q_3}{5}\right] = -54.8\ mJ$$

Electric Energy Density

In the case of continuous charge distributions, defined by a charge density (ρ_L, ρ_s or ρ_v), the energy stored in the electric field is calculated using an integral equation.

Starting with the point charge equation $W_E = \frac{1}{2}\sum q_i V_i$

$$W_E = \frac{1}{2}\int \rho_L \, V \, dl$$ Line charge

$$W_E = \frac{1}{2}\int \rho_s \, V \, dS$$ Surface charge

$$W_E = \frac{1}{2}\int \rho_v \, V \, dv$$ Volume charge

Substituting $\vec{\nabla}\cdot\vec{D} = \rho_v$ and $\vec{E} = -\vec{\nabla}V$ and using the divergence theorem, we obtain

$$W_E = \frac{1}{2}\int (\vec{D}\cdot\vec{E})\, dv = \frac{\varepsilon}{2}\int |\vec{E}|^2 dv \quad \rightarrow \quad w_E = \frac{\varepsilon}{2}|\vec{E}|^2$$

Electrostatic energy stored in the field (J)

Electrostatic energy density (J/m^3)

Q & A

Send me an email if you have any questions.